

LLMs in Wikipedia: Investigating How LLMs Impact Participation in Knowledge Communities

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Large language models (LLMs) are reshaping knowledge production as community members increasingly incorporate them into their contribution workflows. At the same time, participating in knowledge communities involves more than just contributing content - it is also a deeply social process shaped by members' level of expertise. Although communities must carefully consider appropriate and responsible LLM integration, the absence of concrete norms has left individual editors to experiment and navigate LLM use on their own. Understanding how LLMs influence community participation across expertise levels is therefore critical in shaping future norms and supporting effective adoption. To address this gap, we investigated Wikipedia, one of the largest knowledge production communities, to understand participation in three dimensions: 1) how LLMs influence the ways editors gather knowledge, 2) how editors leverage strategies to align LLM outputs with community norms, and 3) how other editors in the community respond to LLM-assisted contributions. Through interviews with 16 Wikipedia editors with different levels of expertise who had used LLMs for their edits, we revealed a participation gap mediated by expertise in adopting LLMs across knowledge gathering, community norm alignment, and peer responses. Based on these findings, we challenge existing models of novice editors' involvement and propose design implications for LLMs to support community engagement, highlighting opportunities for LLMs to sustain mentorship, knowledge transmission, and legitimacy building by scaffolding and feedback, process documentation, and LLM disclosure by good-faith novice editors.

CCS Concepts: • **Human-centered computing** → **Empirical studies in collaborative and social computing**.

Additional Key Words and Phrases: Knowledge Production, Large Language Models, Human-AI Collaboration, Social Computing

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1 Introduction

Knowledge production includes the creation, documentation, maintenance, and communication of content [2]. Participation in knowledge production communities is both epistemic – e.g., requiring attention to factuality and balance – and social [11, 63] – e.g., aligning practices to community norms [6, 12], engaging with other community members [50, 63]. Contributors must gather reliable knowledge, structure it appropriately according to norms, and communicate with the broader community. New members of knowledge communities – e.g., students entering a new field, developers joining an open source project, or novice Wikipedia editors – face the challenge of learning the

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community's epistemic requirements and participation norms. The process of entering communities of practice and gradually moving toward more central roles through observation, practice, and interaction is theorized as Legitimate Peripheral Participation (LPP) [11, 42].

The advent of large language models (LLMs) in generating fluent, well-structured content through natural language prompts [8, 71] offers opportunities for knowledge contributors to enhance their knowledge work. [4, 44, 71, 79]. However, the easily obtained, superficially fluent content produced by LLMs may not meet the epistemic and social requirements of a knowledge community such as accuracy, proper use of evidence, and attribution [34]. This tension certainly arises for Wikipedia [5, 63], where no firm consensus or norms around the use of LLMs have yet developed [5, 63]. As a result, individual editors must interpret, navigate, and experiment with appropriate and responsible usage of LLMs that is not only technically effective, but also socially legitimate within the community.

This raises important research issues: *how* are individuals utilizing LLMs in knowledge work and *what* are the effects of LLMs on their community participation? In this work, we consider participation as a socio-technical process that includes both individual LLM use and community responses to the use. We explore these issues in the context of Wikipedia, which is especially critical due to the observed increase in LLM-generated article content [10] – editors are already using LLMs to contribute to Wikipedia. Our approach is guided by prior work connecting expertise with effectiveness of LLM adoption in programming and learning sciences [7, 14], which suggests that novice and experienced Wikipedia editors may exhibit distinct modes of LLM usage to support their participation. Accordingly, we formulate three specific research questions:

- **RQ1:** How do editors with different levels of expertise use LLMs in *gathering knowledge* to contribute to Wikipedia?
- **RQ2:** How do editors with different levels of expertise *align content* gathered through LLMs with Wikipedia norms?
- **RQ3:** How do other editors in the Wikipedia community *interpret, engage, and respond* to LLM-assisted edits?

Through semi-structured interviews with 16 participants (5 novice, 11 experienced editors) who have used LLMs in editing Wikipedia, we discovered an expertise-based participation gap between editors. Experienced editors enhance their participation through LLMs; they expand the range of their contributions, leverage strategies to align LLM-generated content with community norms, and receive positive responses from other community members. In contrast, while LLMs lower the barrier for novice editors to gather knowledge and enter into Wikipedia, these editors tend to over-rely on LLM-generated content; they have not (yet) developed epistemic skills needed to properly evaluate the content. As a result, their edits are more likely to be rejected by the community, as perceived signs of LLM use function as a social heuristic for evaluating content quality.

Our results revealed a participation paradox for novice editors. Through the lens of Legitimate Peripheral Participation (LPP), we showed that the ease of LLM-assisted content generation enable novices to prematurely achieve more central forms of contribution without the gradual learning typically fostered through feedback, knowledge transmission, and legitimacy-building. Rather than supporting gradual learning, LLMs can short-circuit key social learning mechanisms by enabling novices to perform central tasks, such as content evaluation and norm alignment, before they are socially or epistemically prepared.

We situated our work within broader empirical HCI/CSCW research on Human-AI collaboration and communities of practices [21, 49, 56], making the following contributions:

- We provided an empirical account of how LLMs are differentially adopted by novice and experienced editors, and how these uses are socially interpreted and responded to within a knowledge community;
- We extended LPP by analyzing how LLM-assisted contributions can bypass or disrupt established learning pathways for novices, reshaping mentorship and feedback, knowledge transmission, and legitimacy-building processes;
- We presented design implications for tools that support community participation, emphasizing the need for scaffolding, process transparency, and intent disclosure to foster trust, learning, and legitimate participation for new members.

2 Related Work

2.1 Human AI collaboration in writing

Knowledge production involves drafting, sourcing, formatting and summarizing texts. To contextualize this process, we survey the literature in AI-assisted scientific writing [20, 45, 48], given its close parallel to Wikipedia editing, and identify patterns of human-AI collaboration.

LLM-based tools have been developed and evaluated to assist with various stages of the scientific process. For example, Sparks [28] is implemented to inspire connections between scientific concepts. Several studies [1, 76] evaluate the potential of LLMs to conduct literature reviews. Additionally, Radensky et al. [60] introduce a plan-draft-revise workflow and design a writing assistant for scientists to draft blog posts for their research papers.

Researchers are increasingly integrating large language models (LLMs) into their work, shaping both how they use and perceive these tools. A quantitative analysis of 14 million abstracts of published papers over 14 years reveals that at least 10% of 2024 abstracts were likely LLM-assisted, indicated by the suspicious increase in the usage of certain words [40]. A survey study [46] shows that 81% of 816 researchers already incorporated LLMs into different aspects of their research workflow, including information seeking [35, 96], editing [97] and ideation [58, 59] as the most frequent usage. Users often perceive generative AI not merely as a passive writing aid [36], but an idea generator and active collaborator [20, 91] in content creation. Engagement with LLMs differs significantly depending on the use case. For example, users engage in back and forth conversations with Sparks for translating concepts, while they use it more independently in longer sessions for inspiration and perspective taking [28]. Implicit users who were less specific about their goals tend to search for new ideas, while explicit writers seek precise content to incorporate in their writing [91]. Similarly, ChatGPT is used to support users' information seeking behaviors, rather than completing tasks for them [48]. Writers may modify or directly accept the generated text [92], and doctoral students iteratively interact with GenAI assistants in reading, copying, pasting and shaping content in the writing process [54].

We acknowledge that while scientific writing serve as the entry point, Wikipedia extends beyond scientific writing in practices such as aligning with community norms. Building on this body of prior work, we examine how knowledge contributors use and interact with LLMs in communities with norms. Our findings extend existing research by revealing use cases within the knowledge production processes and identifying similar user behaviors. Most importantly, we find that expertise influences editors' ability to incorporate LLM-generated content in their own writing. As a result, their participation in the knowledge community is affected.

2.2 Knowledge production in Wikipedia

In Wikipedia, participating in the collaborative knowledge work entails three major aspects: 1) contributing content 2) engaging with other editors and 3) enforcing norms.

2.2.1 Content. The central component of knowledge work on Wikipedia focuses on the content or knowledge itself. Rooted in a range of intents such as adding supporting evidence and removing existing information [61], editors perform insertions, deletions, modifications, and relocations [18, 19]. Such adjustments to articles target different elements in Wikipedia articles, including texts, links, references, and templates [18, 19]. Yang et al. [89, 90] categorize edit actions into two larger groups: meaning-preserving edits and meaning-changing edits. While meaning-preserving edits include modifications such as paraphrasing, spelling/grammar, and relocation, meaning-changing edits contain insertions, deletions and some level of modifications. Built on 14-label taxonomy from [89], Ruprecht et al. [66, 67] derive 3 super-labels: content modification to actual information on the article page, format without changing of meaning on text, and WikiContent such as processing tags and vandalism fighting.

Multilingual edits across language editions [30] are important for knowledge production, especially on Wikipedia, because of (1) the nature of more than 300 language editions in Wikipedia [65], and (2) content coverage and language gaps that contributors aim to bridge [62]. As the quality [65] and quantity of articles across Wikipedia language editions vary diversely [43, 88], multilingual editors play an essential role in bridging and filling the gaps from one language edition to another [30, 37, 88].

2.2.2 Community. As editors gain experience on Wikipedia, their focus often shifts from individual content edits to active engagement with the editor community [11]. The need for coordination increases, as social interactions with other editors help establish internal consensus [6] (e.g., structure of the page [63]), resolve conflicts [39, 50], especially when they work on the same articles [63]. Editors engage in social interactions [47] in talk pages and Wikiproject groups [63]. Though a small group of editors participate in article talk page discussions [38], editors use both article talk pages and user talk page [39] for diverse purposes, such as requests or suggestions for editing coordination, requests for information, and references to vandalism [93]. Wikiprojects establish a sense of belonging for editors as they socialize with other similar-minded editors [22, 63].

2.2.3 Policies and guidelines. On top of content and community, an editor is expected to follow community norms. Wikipedia policies and guidelines are community norms that support collaborative work [6] and maintain high quality content and credibility of Wikipedia [57]. These policies define not only the standards for allowed content (e.g., Neutral point of view [81], Verifiability [85], Notability [84]), but also prescribe expected behaviors among community members (e.g., civility, dispute resolution, and no personal attacks) [75]. Butler et al. [12] identified their roles such as signals to external organizations, protecting Wikipedia's reputation and outside attack. They also represent collective efforts to organize, coordinate for consistent and reliable contributions. By easing the difficulties of seeking consensus among editors [6], these rules play an essential role as boundary objects [78] to the daily operation in Wikipedia.

2.2.4 From newcomer to members. Legitimate Peripheral Participation (LPP) [11, 42] states that novices start from low-risk and small tasks, and progressively take on more responsibilities. Built on LPP, situated learning [42, 64] emphasizes that this process from novice to member is social: individuals gain skills and knowledge through interactions with other members in collaborative work. In Wikipedia, novice editors typically begin with low-stake tasks such as fixing errors and improving grammars [11]. Through receiving feedback from other community members, the gradual learning pathway to participation enables novices to gain technical skills, understand community norms, and collaborate with others.

In summary, content, community, and norms inspire and guide our research questions, as we reflect that participation in Wikipedia consists of these aspects. They also serve as a basic

framework to inform our interview questions, in correspondence to knowledge contributors' edits and contribution workflows. Legitimate peripheral participation (LPP) and situated learning offer a lens to interpret our results, which we expand in details in Section 5.

2.3 Tool-assisted editing in Wikipedia

Editors and researchers have developed tools to ease the complexity of contributing to Wikipedia. In this section, we mainly present semi-automated bots, and note more recent AI/ML tools.

2.3.1 Semi-automated bots. Bots play a vital role in Wikipedia's collaborative editing system [23] and have been extensively studied in the literature [17, 25, 27, 77]. Bots are designed to automate repetitive or large-scale tasks [77], and most of the early bots are implemented by scripts. For instance, HostBot invites new editors to socialize in Q&A forums [51, 52]. Huggle and Twinkle fight vandals and unconstructive edits [27]. Rambot adds data into country and city articles [33]. Zheng et al. [95] classified bots' roles into 9 categories: generator, fixer, connector, tagger, clerk, archiver, protector, advisor, and notifier.

While bots protect and patrol high quality knowledge in Wikipedia [25], they can also cause unintended consequences. In particular, vandal fighter bots negatively impact newcomers' retention, as they revert contributions from good faith editors that may appear suspicious [32, 95]. This reflects a broader dynamic in human bot collaboration. Editors not only develop, approve [26], operate, and maintain bots [63, 95], but are also influenced by bots' behaviors and work alongside them. For example, human editors check and correct the information by bots [55]. In the context of vandal fighting, bots detect potential vandals and put them in queues. Editors can then decide and authorize bots to revert edits and leave warning messages. In more severe cases, bots can notify administrators when users accumulate multiple warnings [27].

Because bots closely support the editing process, they are generally well accepted [16]. In fact, they are not viewed as simple tools, but social actors [24] that shape collaborative outcomes. Through the evolving collaboration dynamics, humans and bots maintain the values and social structures of Wikipedia together [27].

Notably, emerging AI/ML powered tools are developed not only to assist content editing (e.g., multilingual contributions [9, 41, 43]), but also to support tasks related to editing. For example, SuggestBot [17] and other recommendation systems [88] make edit recommendations to editors. ORES [31] evaluates the quality of articles. Edisum [68] generates informative edit summaries. Wikimedia Foundation recognizes the potential of AI tools, and is developing strategies to continue investing in AI tools that support the editing process and improve editors' experience [80].

Building on prior work, we identify interaction patterns that differ from traditional bot usage. Unlike traditional tools that conform to Wikipedia norms, LLMs bypass policies and guidelines and require editors to make normative decisions, which challenge novice editors' participation.

3 Methods

The researchers collaboratively designed a semi-structured interview protocol, which included 3 sections: 1) general context about contributions, 2) experience of using LLMs in Wikipedia, and 3) vision for human AI collaboration in knowledge production. After receiving an exemption from the Institutional Review Board at the University, the first author conducted pilot studies with two graduate students and refined the interview questions. Data from pilot studies were excluded from data analysis.

ID	Tenure	# of Edits	Fre- quency	LLM models	Gender	Age	Race/Ethnicity
P01	0-2 years	~100	monthly	ChatGPT, Gemini, You.com	Male	35-44	White/Caucasian
P02	2-5 years	10k+	daily	ChatGPT			
P03	5+ years	21k+	daily	ChatGPT, Grok	Male	55+	White/Caucasian
P04	5+ years	2k+	weekly	ChatGPT		25-34	White/Caucasian
P05	5+ years	50k+	daily	ChatGPT	Male	25-34	White/Caucasian
P06	5+ years	1k+	weekly	ChatGPT	Male	35-44	Avropoid- Caucasian/Azer- baijanian turkish
P07	5+ years	159K+	daily	ChatGPT	Male	45-54	Middle East- ern/Arab
P08	5+ years	12k+	daily	ChatGPT	Non-binary	18-24	Asian
P09	0-2 years	5.3K+	daily	ChatGPT; Claude	Male	25-34	White/Caucasian
P10	2-5 years		monthly	ChatGPT	Male	18-24	Black or African American
P11	5+ years	~10	yearly	ChatGPT			
P12	5+ years	55k+	weekly	ChatGPT; Claude; Llama	Prefer not to answer	35-44	White/Caucasian
P13	2-5 years	4k+	daily	ChatGPT; Grok	Male	18-24	Bengali
P14	0-2 years	~200					
P15	0-2 years	~200					
P16	5+ years	83k+	monthly	ChatGPT	Male	35-44	White/Caucasian

Table 1. Participant summary, adopted from [72]. Some fields are left empty due to incomplete responses or participants’ preference to not answer.

3.1 Recruitment and Participants

We followed Wikipedia’s convention [87] by defining novice editors (newcomers) as those with fewer than 200 edits and experienced editors as those with more. We recruited novice and experienced editors as participants through multiple channels from English Wikipedia. We posted our interview invitation on the WikiMedia project page. In parallel, we searched for editors who had potentially used LLMs and sent invitations to their talk pages or via email if they had enabled that feature. We also leveraged snowball sampling [53], asking for referrals from our participants. Editors who were interested in the interview study filled out a survey about basic usage of LLMs, experience on Wikipedia, contact and demographic information based on standard practices. We

then sent emails to schedule the interview on Zoom, along with a consent form. One participant responded to our interview-scheduling email but ultimately declined due to time constraints. In addition, several participants indicated discomfort with video/audio recording. Thus, we offered the option to participate in the interview through written text (email).

As a result, we conducted 16 interviews, of which 5 were conducted via email. We began to observe data saturation [69] after finishing the 13th interview. Thus, we stopped recruiting more participants after completing 16 interviews, which was consistent with the average sample size for qualitative research at a top HCI conference [13]. Participants did not receive monetary compensation, and their demographic information along with their expertise in Wikipedia is listed in Table 1.

3.2 Interview Procedure

Consent was acquired from each participant before the interview began. Before the interview, we thanked them for their participation and noted the privacy considerations - the interview would be recorded only for data analysis purposes; video was not required; participation would remain anonymous. As each participant fully understood and agreed, we began the interview and followed the interview protocol.

First, we briefly introduced the goal of the study. We then asked participants about their contribution process such as typical tasks, general workflow, and common practices or guidelines to understand the context. Next, we asked participants to think of a specific use case of LLMs in their contribution, and asked relevant questions such as interactions with LLMs. Then, we asked them to envision the future for human AI collaboration in knowledge work and ideal interactions with LLMs. We concluded the interviews by inviting participants to share any additional information. In the semi-structured interview, we allowed participants to freely share their experiences and asked follow-up questions.

3.3 Data Analysis

Our 11 recorded interviews ranged from 42 minutes 22 seconds to 1 hour 17 minutes 29 seconds, with an average of 1 hour 3 minutes 18 seconds. After transcribing the interviews, we conducted an inductive thematic analysis approach [15]. We used ATLAS.ti, a popular software for qualitative data analysis. First, the authors open coded three interviews together to establish a shared standard for open coding. Then, we asynchronously open coded the remaining interviews. We ended up with 1524 codes, for example, “P05 - LLMs has no sources”.

After that, the authors collaboratively grouped the codes based on their similarity and relevance in Miro. Themes and sub-categories emerged through this iterative process, and the authors discussed and reached agreement for the name of the themes and sub-categories. For example, the code mentioned above was put under the sub-category “no sources,” and the theme “violates Wikipedia policies.”

4 Results

Our results surfaced how Wikipedia editors leveraged and interacted with generative AI (see Table 2 for LLM use cases). We identified three aspects of the knowledge production process affected by LLM usage: gathering knowledge to contribute to Wikipedia, aligning that knowledge with Wikipedia norms and policies, and peer response to editors who used LLMs. Across these dimensions, the experience of editors varied significantly based on their level of expertise in Wikipedia.

4.1 RQ1: Gathering and contributing Knowledge

Our findings for RQ1 showed that editors’ expertise levels are associated with distinct patterns of LLM use in content contribution. For experienced editors, LLMs extend cognitive capacities,

Category	Use cases	Highlights
Generate	Generating examples	<i>Generating practical examples for theoretical concerns. (P14)</i> <i>I've been learning. So, I need a guide... If there is anything that could guide me for better content. (P15)</i>
	Providing editing guidance	
	Creating articles	
	Writing codes for articles	
Search	Searching sources	<i>I think it's better at searching for those than I am...it is a skill to phrase your queries in Google and know how to find things. (P04)</i>
	Suggesting images	
	Searching information	
Refine	Copyediting	<i>Google cannot do it. DuckDuckGo cannot do it. But language models are quite often able to do it well, and give me the short name of a thing. (P07)</i>
	Formatting	<i>In the old days, I would have copied this thing, put it into a plain text file, changed it to a CSV, uploaded it into R, and written a script. But with ChatGPT, I can just post the whole thing to ChatGPT, say I want it in this format. (P04)</i>
	Translating	<i>You use wikitext, like square brackets to link to a page and you use single quotation marks to make something bold or italic and all this. Google Translate does not understand this wikitext language, but ChatGPT does. (P05)</i>

Table 2. Selected LLM Use Cases Reported by Participants. *The distinction between generation and refinement lies in whether the content exists prior to interacting with the LLMs. In generation, LLMs are used to produce new text from scratch. In refinement, editors input existing content to receive feedback or suggestions for improvement. Copyediting includes grammatical improvements, rephrasing suggestions for repetitive words or longer description, brainstorming alternative expressions.* The shading aims to improve readability.

enabling exploration of new topics and improving confidence and quality. In contrast, novice editors often rely on LLMs to fill in their knowledge gaps, as LLMs reduce the barriers to entry and provide a sense of guidance.

4.1.1 Experienced editors venture into new topics and tasks. LLMs have the ability to guide editors toward content they wouldn't have otherwise considered, by providing useful contextual information without requiring proactive efforts from the editors. P05 shared: “*The articles that I created using AI, I probably would not have created them [by myself].*” This is especially true when available resources are limited. P08, who was writing code for supplementary graphs to improve article quality, didn't know anyone to consult to troubleshoot his R code, especially as it might need a long time to fix the bugs. LLMs served as a supportive tutor, as he reflected:

“[ChatGPT] is basically the next closest thing to [people who can help]. It brings up different packages that I wouldn't consider...[LLM] definitely directed me to things that I would have never been able to think of on my own.”

In this way, LLMs enabled him to engage in tasks he would not have attempted alone. P03 shared a similar anecdote. As he went down a rabbit hole on fruits and vegetables, LLMs suggested a new section he didn't even know about and would not think of:

“[ChatGPT] suggested a section where I didn’t even know that a specific vegetable had been - a hybrid of this vegetable created another one that was not in the article...I looked to make sure the information was factually correct, and I put that in there.”

For P03, he explored novel topics besides his typical contributions in business articles. The capacity of LLMs to introduce new, previously overlooked knowledge enhances editors’ creativity and encourages them to venture beyond their usual domains, most importantly, add the piece of knowledge in Wikipedia.

4.1.2 Experienced editors are exposed to new perspectives. In addition to helping editors move beyond their typical domains, LLMs extend cognitive capacity by introducing unexpected and insightful perspectives. P08 recounted LLMs supporting him to reflect on his own text in a different way, for instance, by suggesting what is interesting about his text:

“I’m asking LLMs for ideas...“What’s interesting in this big block of text I’ve written?” Honestly, I spent 48 hours staring at my computer screen writing this article. I’ve kind of bored myself to death with it.”

LLMs could also serve as a simulated average reader, offering pluralistic perspectives that editors find helpful in reviewing their content. P09 noted, *“I regularly ask LLMs...when I want a pluralistic opinion. You want to know what the average reader would think.”*

4.1.3 Experienced editors are more confident in unfamiliar areas. This cognitive support, in turn, increases confidence and satisfaction for experienced editors, even in domains they are less familiar with. P03 felt more capable of editing complex topics such as medical and legal content: *“I can digest the information if I don’t understand...and run through LLMs to simplify for myself.”* P04 treated LLMs as an additional validation mechanism: *“LLMs are a good extra filter to make sure there’s not something you overlook or mistakes you are making.”* P08 remembered editing without LLMs took endless iterations, as he would *“write, write, try to get through it,”* especially when he got stuck. The presence of LLMs helped him feel satisfied with his work sooner, even improving his sleep routine: *“it allows me to feel happy with what I’ve written earlier...thus allow me to go to bed earlier.”*

4.1.4 LLMs lower entry barriers for novice editors. Novice editors themselves frequently shared that LLMs encouraged them to edit Wikipedia. For example, P14 described the challenges related to information access in his local context, and how LLMs helped with research that would otherwise take days:

“LLMs encourages me to work on Wikipedia...I have to go to libraries to look for information...I must pay for the transportation and a lot of other things...Before I go to the library, I’ll check on AI to know more about the topic.” (P14)

P01 found that proofreading using LLMs made editing easier, particularly given that English was not his native language: *“English is not my native language...I have to proofread it on ChatGPT.”* More experienced editors agreed that LLMs lowered the barriers for entry, as P12, P09, P04 and P05 pointed out that LLMs *“increase the ability of new contributors to contribute, as it breaks down a lot of the barriers (P12).”*

4.1.5 Novice editors rely on LLMs to fill in their knowledge gaps. As LLMs are able to lower the technical and linguistic barriers, newcomers tend to rely on LLMs to fill in their knowledge gaps, often in generating and searching for content. Editors observed that newcomers might use LLMs to generate content. P04 observed that many newcomers were using LLMs to draft articles: *“There’s an increased expansion...the vast majority of them just write a prompt like, hey, ChatGPT, can you write a Wikipedia article on this?”* Newcomers also leveraged LLMs to search for knowledge around the subject topics, and specific wiki knowledge. For instance, P10 shared that in order to ensure

high quality contribution, he “use[d] LLMs as my research platform to know more about what I’m looking for.” P01 emphasized that he used ChatGPT to “increase my knowledge around the subjects.” He described LLMs as a wise mentor he could learn from.:

“I take the information and integrate it into my thoughts, like speaking with a wise person..I don’t have the luxury of speaking with super intellectuals around myself...I compensate for the gap of knowledge.” (P01)

Similarly, P14 perceived LLMs to be a “school” for him, specifying that LLMs allowed him to “know more about anything inside my house.” He demonstrated this by sharing an example of him deciding to create an article on English Wikipedia, and asking LLMs “how do I start doing this?” and learned that the articles needed “citations, notability and other things.” In this way, LLMs can be a guide for newcomers. P15 reasoned that “I need a guide...If there is anything that could guide me for better content.”

4.2 RQ2: Aligning with community norms

To answer **RQ2**, our findings showed that editors rely on core content policies and guidelines to navigate the unclear norms around LLM usage. They recognize that LLMs could violate these standards, indicating that LLM use is situated within the Wikipedia editing ecosystem. Experienced editors described three strategies to make editorial judgments: evaluation, verification, and modification. As a result, LLMs challenged novice editors, as they had not yet developed skills to make complex judgments and align to community standards.

4.2.1 LLMs fail to align with core Wikipedia content policies. Participants consistently described limitations of LLMs from the lens of violating core Wikipedia policies, as P02 summarized, “LLM output has many flaws and often violates Wikipedia policies.” The most frequently mentioned concerns were related to the Neutral Point of View (NPOV) [82], Verifiability [86], and No Original Research (NOR) [83].

NPOV [82] is defined as “representing all the significant views on a topic fairly and proportionately without editorial bias.” LLMs often produce language that violates NPOV due to an overly positive or promotional tone. P05 noticed that LLMs tended to use phrases such as “it’s one of the best” or “there are so many possibilities,” which are discouraged unless supported by reliable citations. He added that such content often contained “puffery peacock terms,” especially when asking LLMs to write something from scratch. P07 agreed with P05, as he observed that LLMs could use English expressions not typically found in Wikipedia. This aligns with prior work [3], which found that LLMs are limited in detecting and generating content that conforms to NPOV like a community expert.

Verifiability [86] refers to the ability of a claim to be proven right or wrong, typically through the presence or availability of sources. It is compromised when LLMs cite unreliable sources or fabricated sources. P07 criticized a LLM named Perplexity. While sometimes it “[didn’t] give any sources,” other times when it gave sources, they would not meet the Wikipedia standards, for being “unreliable junk commercial websites that are made not for providing correct or verified information, but just for gathering clicks and showing advertisements.”

LLMs are prone to hallucination, especially for obscure content. This violates NOR [83] which prohibits original research that finds no reliable or published sources to support. More importantly, LLMs’ tendency to hallucinate contradicts the very core value of knowledge integrity. P04 specified that LLM-generated content was “not necessarily accurate” as LLMs “can hallucinate and make up sources.” P02 and P03 also mentioned hallucination, as P02 explained “when the model invents non-existing facts and invents non-existing references to support claims.” P08 shared specifically about

LLMs struggling with less common programming languages, noting their limited performance beyond languages like Python or R.

4.2.2 Experienced editors' strategies to make editorial judgements. To cope with the risk that LLMs may violate policies and guidelines, our participants emphasized their responsibility as gatekeepers. P03 stressed, “*even if you have all the tools available, you are still the last one to make the decision whether this should go in or not.*” In practice, editors used three key strategies when working with LLM-generated content: evaluation, verification, and modification.

Evaluation involved assessing whether LLM suggestions improved the readability and coherence of the content. P08 put it as “*make judgement about whether the LLM content makes sense.*” P02 developed a practice of “*evaluating whether any of the [suggestions] would improve the writing.*” P09 evaluated the quality in terms of consistency as he “*regenerated the answers to see if it's consistent, [and] compared the answers of multiple chatbots.*”

While evaluation ensures readability and coherence, verification checks the content's factual accuracy and reliability. P09 and P04 both stressed “*verifying that everything makes sense and is supported by sources (P09).*” P07 noted the necessity to “*check every word that LLM wrote because of verifiability.*” He mentioned specific strategies for verification: reaching out to friends with questions, and cross-referencing terminologies or phrases via web searches.

Additionally, participants intensively modified the outputs before incorporating them into Wikipedia to align with Wikipedia's tone and style. P16 noticed that LLMs tended to be overall more positive, and depicted modification as neutralizing the tone of the text. P07 would “*heavily edit*” the output by himself before publishing. P03, P04, and P05 would modify the text to better fit the context, such as “*wiki format (P04)*” the links and fixing typos in wikitext, as “*ChatGPT could translate the wikilinks or reference wrongly so I have to update and change them (P05).*”

4.2.3 Newcomers as gatekeepers? Being gatekeepers for LLM-generated content requires expertise and wiki knowledge, specifically around Wikipedia norms and standards. However, new editors have not yet developed these skills. For example, newcomers recognized that they did not know all the guidelines and principles as P01 pointed out “*I am still developing my skills. I'm new to Wikipedia, so I may not know all the guidelines and principles.*” P03, an experienced editor, talked about how he made decisions when interacting with LLMs: “*I am the big decider of what I know in my gut what should be in Wikipedia.*” And he pointed out that editors with many years of contributions would have the expertise to tell whether materials should be in Wikipedia or not. He then contrasted editors with different levels of expertise: “*I have 20 years of editing right now. So, the tool is much more of an enhancement to me because I know what's allowed and not allowed quickly. I can accept or reject a suggested edit...[but] a new user would go through and just accept, accept, accept the edit.*” P02 mentioned that “*inexperienced editors often lack the ability to evaluate whether an output is appropriate...This creates problems when inexperienced editors unaware of the limitations of LLMs rely on them to make contributions without critical examination.*” This suggests that newcomers lack the ability to critically evaluate and examine LLM-generated content and may over-rely on LLM outputs, which leads to challenges in producing high-quality content. This is well documented in an example P15, as a newcomer himself, shared:

“I was trying to get good content out of it, but it had so many unwanted things as well. As I'm learning, I'm not that clear about the content exactly, what should I filter out. Sometimes even if I read it multiple times, I do not know whether it is needed or not, or whether the content is good or not, if this particular explanation is useful or not. Then after this editor pointed it out, I noticed that's not what [Wikipedia] wanted.” (P15)

4.3 RQ3: Community response

Our results showed that other Wikipedia community members respond differently to LLM-assisted edits made by novice editors compared with those made by experienced editors. Other editors praise and accept LLM-assisted contributions from experienced editors, yet call out and reject those made by novices. LLM usage becomes a new evaluation heuristic for community members, as vigilant community members would read low-quality content as “AI automation” and reject the content.

4.3.1 Other community members respond to experienced editors with approval. Experienced editors often received positive feedback. P16 shared that one editor praised him for copyediting a section this editor had written, “*better than he had ever seen before.*” He clarified in the interview: “*it was chatgpt that had done it.*” P04 also shared that his LLM-based edits were well received: “*They’re not really received any differently than in my non-LLM based edits.*” P07 also noted that no one noticed or identified that he had used LLMs, due to extensive post-editing.

4.3.2 Other community members respond to newcomers with rejections. In contrast, novice editors faced heightened scrutiny because other editors easily detected that their edits result from LLM usage. For example, P01 was told his contribution was “*machine generated,*” and P10 was criticized for making “*promotional and essay-like articles.*” P15 shared that another editor quickly identified his use of ChatGPT, which prompted him to remove all LLM-generated content.

4.3.3 Perceived LLM use as an evaluative signal in community review. Perceived signs of LLM use became part of community evaluations, particularly when contributions were seen as low quality or norm-misaligned. Several newcomer participants stated that their drafts were quickly flagged as “AI-generated” and rejected; interestingly, most of the times the rejection made no reference to specific content issues or violations. P11 recalled that after he pushed his article out as a draft, he “*got a message from somebody in the community saying, ‘oh, it looks like you created this with an LLM and we don’t want that kind of material.’*” P05 noted that some editors “*associated poor content with AI*” and that content believed to involve LLMs “*got removed pretty quickly.*” These accounts suggest that noticeable LLM use in and of itself is treated as a signal of a low quality edit.

Not all LLM-assisted edits were inherently low quality; that is, concerns about LLM usage went beyond clearly problematic contributions. P09 brought up an observation of editors attacking good edits because they were perceived to be generated by LLMs: “*I have once seen spiteful comments...other people severely criticized [the article] for being AI-generated.*” In other words, the community care not only about the actual quality of of an edit, but also about how it was produced. Perceived LLM use undermines trust: it suggests that an editor might not have understood their edit and verified the content. As P04 noted: “*it’s so ambiguous...people might assume that [LLM assistance] means you copy and pasted text from an LLM prompt without any review or understanding.*” This helps clarify why “obvious” AI edits are greeted with suspicion and rejection. And on the other hand, experienced editors know how to edit any content they generate with an LLM to remove obvious “AI markers”, which explains why their AI-assisted edits are accepted.

5 Discussion

5.1 The paradox of participation

Our findings revealed a paradox of participation: While LLMs lower entry barriers for newcomers, this can result in these editors producing low-quality content due to unfamiliarity with community norms; even more simply, if they don’t know how to edit LLM-assisted content to remove “generated with AI” markers, even acceptable edits can be rejected by the community. Moreover, since noticeable markers of AI content are treated with suspicion, even potentially acceptable edits can

receive negative feedback, thus exacerbating the challenges that newcomers face. [32, 52, 74]. We interpret this paradox using the lens of Legitimate Peripheral Participation (LPP) [11, 42, 42, 64], which proceeds from the insight that becoming a true participant in a community involves a social learning process of mentorship and feedback, knowledge transmission, and legitimacy building.

5.1.1 Mentorship and feedback. LPP theorizes that newcomers gain competence by interacting with experienced community members and receiving iterative and situated feedback. In the context of Wikipedia, this process typically follows a gradual path [11]: newcomers may begin with peripheral tasks such as fixing typos or formatting templates, then move toward harder and more central tasks such as making large edits to an article and even creating new articles. New editors can get feedback in talk page discussions or edit summaries, helping them learn about Wikipedia policies, norms, and practices. Our findings suggest that LLM usage for Wikipedia may reshape this learning pathway in two ways: 1) skipping peripheral tasks, and 2) reducing opportunities to learn from constructive feedback.

First, LLMs may bypass early-stage mentorship by enabling newcomers to immediately generate content that would traditionally require incremental learning and guidance. Several newcomers describe using LLMs to draft or create articles; in other words, they go quickly to central Wikipedia tasks, short-circuiting the process of making small edits and receiving feedback and mentorship from experienced editor. Therefore, while using LLMs makes it easy to produce *an* edit, these editors skip the steps that make it possible to produce *acceptable* edits.

Second, LLM use appears to change the nature of feedback newcomers receive on their edits. Our findings indicate that edits perceived as AI-generated are more likely to attract flags, critiques, or removals that focus on the use of LLMs itself, rather than on the quality of the content or specific policy violations. This means newcomers are less likely to get explanations in terms of Wikipedia policies and norms, such as verifiability and neutrality, and constructive guidance concerning how to produce acceptable edits. Thus, instead of a newcomer's edit – even if unacceptable – resulting in a learning opportunity, it triggers plain rejection.

5.1.2 Knowledge transmission. LPP describes newcomers as developing competence not only by contributing and receiving mentorship and feedback, but also by observing how experienced members perform work, make judgments, and align their actions with community norms [11, 42]. In Wikipedia, this includes observing how experienced editors argue, negotiate, correct errors, and respond to others in talk pages and edit histories. Such situated knowledge is transmitted through exposure to these moments of practice-in-action. Our findings suggest that the introduction of LLMs into the editing processes possibly may interrupt this knowledge transmission, as newcomers have no visibility into experts' LLM practices.

Newcomers have no visibility into expert practices of LLM-assisted editing. In our interviews, experienced editors describe engaging in extensive evaluation, verification, and modification when working with LLM-generated content. However, these deliberative processes are typically invisible to newcomers – expert editors do not 'advertise' the fact *that* they used AI nor expose *how* they used it to produce acceptable edits. Newcomers see *what* a policy-compliant contribution looks like, but not *how* experienced editors arrive there. This reduces the opportunities for observational learning that are central in a legitimate peripheral participation process.

5.1.3 Legitimacy building. In LPP, legitimacy is socially constructed through interactions. Newcomers earn trust as their contributions are interpreted as good-faith efforts, even when they are incomplete or imperfect [42]. In Wikipedia, experienced community members differentiate between learning mistakes and disruptive vandalism [70]. Our results indicate that the introduction of LLMs complicates this process by obscuring the social signals that support legitimacy-building.

Specifically, LLM use disrupts how community members interpret mistakes and assess contributor intent. Traditionally, low-quality content was evaluated through one of two frames: good-faith newcomers' learning mistakes that could be addressed through constructive feedback, or vandalism that required removal. However, our findings suggest that edits suspected to involve LLMs introduce a third, ambiguous interpretive category. Community members evaluate such edits not only based on content quality, but also based on their perceived origin — with obvious LLM use becoming a signal of *illegitimacy* that reduces trust in both the edit *and the editor*. We speculate that this ambiguity arises because LLMs entangle effort and intent: AI-assisted edits obscure how much work, understanding, or judgment an editor has invested, making it difficult for community members to infer whether a mistake reflects a good-faith attempt to learn or something more problematic. As a result, when newcomers use LLMs, their intent becomes harder to interpret, weakening a key social signal that previously supported legitimacy-building during early participation.

5.1.4 Summary. Viewing our results through the lens of Legitimate Peripheral Participation reveals how the use of LLMs in Wikipedia impacts participation and learning in Wikipedia. Newcomers may face reduced opportunities for constructive feedback and learning by observing experts, and their use of LLMs may brand their efforts as illegitimate rather than good faith attempts to learn.

5.2 Design implications

A key insight of our results is that LLMs *empower* editors, but *undermine* learning and legitimacy. *This tension is likely to persist even as LLMs rapidly improve in capability.* Therefore, we believe that LLMs should be explicitly designed to *support human Wikipedia editors* rather than automating editing tasks. Human editors must retain responsibility for judgment and verification; they must be able to “do the work” of ensuring edits meet Wikipedia policies and norms so that they can “show their work” as necessary. In this section, we outline three design implications that balance the potentials and challenges of LLM adoption in online knowledge communities.

5.2.1 Restore and strengthen mentorship. LLMs can support mentorship by acting as in-process facilitators that guide newcomers and prompt reflection during the editing process, before content is submitted to Wikipedia. Rather than producing immediate article text, LLMs can scaffold participation by decomposing complex tasks into incremental steps and prompting editors to actively engage with key knowledge practices. For example, when a newcomer asks an LLM to generate a Wikipedia article, the system could instead guide the editor to identify relevant sources, summarize claims from those sources, and structure the article by asking questions such as “Which sources would you consider for this topic?” This shifts the LLM from author to facilitator, preserving opportunities for learning through doing.

At the same time, LLMs can incorporate reflective feedback grounded in community norms. By posing questions from the perspective of experienced editors (e.g., “Does this claim satisfy the Verifiability requirement?”), an LLM can encourage newcomers to pause and reflect on policy alignment before publication. Both approach would rely on prompt engineering and retrieval augmented generation grounded in Wikipedia policies and reflective questions [73, 94], which is technically achievable and aligns with existing community norms without changes to Wikipedia or LLM infrastructure.

5.2.2 Support transparent knowledge transmission. We suggest a “Show my work” mode for experienced editors. A LLM-based system can document prompts, revisions, and post-editing decisions an editor made during content creation. Editors could choose to publish these documents on talk pages, allowing other editors – newcomers in particular – to observe how sources, policies, and judgment are integrated into final outputs. While this feature requires careful design around privacy and

consent, it is technically feasible especially given that agentic LLMs are capable of leveraging rich interaction histories to support complex, multi-step reasoning and documentation.

5.2.3 *Rebuild legitimacy pathways.* Another step to support newcomers is to make their good-faith efforts, including reasonable use of LLMs, more visible. For examples, editors could tag their LLM-assisted drafts with specific intentions (e.g., “searching sources,” “copyediting”). These tags, grounded in empirical use cases from our study (see Table 2), can signal status of an edit and editor intent, helping the Wikipedia community identify good-faith efforts.

Beyond disclosure, communities can gradually shift evaluation toward content quality and policy adherence rather than content origin. Moderation tools could surface indicators such as citation completeness, neutrality, and source quality alongside disclosure tags. Complementing this, communities can curate structured feedback template that enable experienced editors to provide consistent, policy-grounded guidance with reduced cognitive burden. Both designs do not require LLMs themselves, but template support and community consensus, making them achievable within existing Wikipedia governance structures.

5.2.4 *Summary.* We presented three design implications for future LLM-based assistants in knowledge production communities. These implications align with the WikiMedia Foundation’s strategy regarding AI tools for editors [80], particularly in terms of helping newcomers understand policies and provide feedback on their edits. Our implications respect and are grounded in the traditional learning pathways described by Legitimate Peripheral Participation (LPP). Therefore, we recommend designing assistants to guide newcomers through the mentorship, knowledge transmission, and legitimacy building. However, our results hint at reality where newcomers continue to use off-the-shelf LLMs to bypass peripheral tasks and begin attempting significant edits immediately. This raises a question: if this comes to pass, how should the community and researchers respond? Future research is needed to explore this question.

5.3 Limitations and future work

Our study provided a in-depth, theory-informed understanding of how Wikipedia editors interact with LLMs. We surfaced empirical findings around interaction patterns, tensions, and community dynamics. However, we acknowledge several limitations.

First, while our qualitative approach let us understand nuanced editor experiences and perceptions, it did not quantify the prevalence of observed phenomena or estimate effect sizes of LLM-related impacts. Our analysis did not aim for statistical generalization; rather, it identified recurring use cases, interaction patterns, and participation mechanisms that can inform subsequent quantitative study. For example, future research could build on our use cases (see Table 2 for LLM use cases) to design surveys or large-scale data analysis to assess the prevalence and distribution of LLM usage, and use our empirical findings and interpretive lens (e.g., the participation paradox) to inform system designs and contextualize quantitative results.

Second, our recruitment methods reflected our focus on editors who have used LLMs in their editing process, which met our goal of investigating users’ actual experiences. However, since our participants were self selected, this might result in bias towards editors with stronger motivation to contribute and higher curiosity and confidence levels for technology adoptions. Most important, it may have omitted editors who deliberately chose to not engage with LLMs. Future studies should include resisting and disengaged editors, especially those who initially chose to use LLMs but later stopped.

Third, our study focused on English Wikipedia, and all participants primarily contributed to English articles. We decided this scope in part because LLM adoption is most prevalent in English at the time of our study. As a result, our findings may not directly transfer to non-English Wikipedia

communities, where LLM availability, capabilities, participation norms and dynamics may differ. Separately, our findings about the uses of LLMs in translation may inspire future work to examine LLM-assisted editing from a multilingual perspective. In addition, although we reported standard demographic information (e.g., race), we did not systematically collect participants' native language or geographic regions. This limits our ability to examine how language capabilities or culture shape editors' experiences with LLMs. Future studies should explicitly take these dimensions into account.

6 Conclusion

To address the gap in understanding the impact of adopting generative AI on knowledge contributors' participation in communities of practice, we recruited 16 Wikipedia editors who had used LLMs in their editing process. Through semi-structured interviews, we examined how editors perceive, adopt, and interact with LLMs. Our findings revealed that LLM adoption introduced a participation gap between novice and experienced editors, manifested in differences in knowledge gathering and contribution, alignment to norms, and most importantly, social responses from the broader community. For novices, we identified a participation paradox in which LLMs simultaneously lower barriers yet undermine learning trajectories. In contrast, experienced editors have the knowledge to benefit from LLMs and enhance their participation, by exploring new topics, understanding diverse perspectives, and gaining confidence in unfamiliar domains. These asymmetric outcomes suggested that LLMs may amplify existing inequalities in participation, echoing "rich-get-richer" dynamics observed in prior studies [29]. Beyond participation dynamics, our findings also raised broader concerns about the use of LLMs for content generation on Wikipedia. For example, editors expressed skepticism toward LLM-generated content due to hallucinations, difficulties with verification, and loss of diversity. Together, our results suggested that appropriate uses of LLMs on Wikipedia are not evaluated solely by efficiency or quantity. Instead, designers and communities must consider how LLMs affect learning, trust, and participation by restoring mentorship through scaffolding, supporting transparent knowledge transmission, and rebuilding legitimate pathways. Our study demonstrates the importance of user experience research in shaping equitable AI integration in communities of practice, and highlights future opportunities for designing AI-powered tools that not only support production but also foster community participation and collaboration.

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